

## XAVIER CHEN

La Jolla, California 92092, +1 (341)333-9519, chenpy2000@gmail.com, Github: chenpy2000

### RESEARCH INTERESTS

---

Language models and machine self-cognition, inspired by principles from neuroscience. Interested in exploring hierarchical architectures for language models to develop autonomous intelligent agents (decision-making, emotion, memory, learning, and hormonal regulation from neuroscience) and tuning language models to align with the reasoning patterns and behavioral traits of specific individuals.

### EDUCATION

---

**University of California, San Diego**, La Jolla, United States Expected Graduation in Jun. 2026

Master of Science in Computer Science

- GPA: 3.93/4.00 for AI, 4.00/4.00 for Neuroscience
- Coursework in AI: Information Theory, Random Process, Machine Learning, Deep Learning
- Coursework in Neuroscience: Systems/Computational/Cognitive Neuroscience, Neurobiology Laboratory

**Tongji University**, Shanghai, China

Aug. 2018 - Jul. 2023

Bachelor of Engineering in Vehicle Engineering (Automobile), Minor in Artificial Intelligence

- GPA: 4.34/5.00, 88.4%. Equal to 3.84/4.00 in US system

### CERTIFICATIONS

---

**Google Cloud**, Professional Machine Learning Engineer Certification Aug. 2025 – Aug. 2027

**MIT xPRO**, Online, Professional Certificate in Data Engineering

Jul. 2024

### RESEARCH EXPERIENCE

---

**Cottrell Lab, UC San Diego, La Jolla, CA, US**

Feb. 2025 – Jun. 2025

*Researcher*

Advisor: Prof. Garrison Cottrell, Department of Computer Science & Engineering, UC San Diego

- Research the parallels between feature representations in artificial neural networks and neural activity in brain regions. Specifically, investigate how different neural network architectures impact identity classification accuracy for both upright and inverted face stimuli.
- Designed and implemented an attractor network model with an innovative feedback loop, enabling the storage of  $32 \times 32$  binary pixel images at a capacity of 400—significantly surpassing the theoretical limit of approximately 150 in a standard Hopfield network.
- Implemented a convolution-based attractor network block that significantly reduced test accuracy on inverted faces to near chance levels, demonstrating a strong ability to constrain representations within the latent space.
- Independently reimplemented the project in a streamlined, maintainable codebase. Significantly accelerates further implementations, ensuring scalability and ability of running across platforms, including local environment, servers and National Research Platform's Kubernetes cluster.

**Tongji University, Shanghai, China**

Apr. 2023 – Jul. 2023

*Research Assistant*

Advisor: Prof. He Lianghua, College of Electronic and Information Engineering, Tongji University

- Investigated how effectively EEG signals reflect human cognition by presenting participants with scenic images and corresponding native-language descriptions across four distinct categories.
- Designed and executed a cognitive neuroscience experiment using a 64-channel EEG cap to capture neural responses under varied contextual stimuli.
- Preprocessed EEG data with the MNE-Python library and co-developed EEG-NET, a neural network model for classifying EEG segments associated with different cognitive scenarios.

**Institute of Fuel Spray and Combustion, Tongji University, Shanghai, China**

Dec. 2022 – Jul. 2023

*Undergraduate Dissertation*

Advisor: Prof. Li Liguang, School of Automotive Studies, Tongji University

- Constructed a test stand for an argon cycle hydrogen direct injection internal combustion engine to explore hydrogen utilization in sustainable energy applications.
- Conducted over 100 controlled engine bench tests under safety regulations to identify optimal control parameters that maximize hydrogen efficiency while maintaining sufficient output power.
- Independently mastered and utilized GT-Suite Simulation Software, employing it to investigate global minimum point and Boundary Conditions leading to engine stalling, systematically analyzing the effects of inlet pressure, inlet temperature, and the ratio of argon to oxygen on engine performance.

**Technical University of Munich, Munich, Germany**

Apr. 2022 – Sep. 2022

*Graduate level researcher*

Advisor: Prof. Daniel Renjewski, Department of Mechanical Engineering, Technical University of Munich

- Modeled bipedal robots in MATLAB, processing from mass-spring representations to 6-degree-of-freedom dynamic models.
- Developed and implemented an Untrainable Neural Network for locomotion control, applying Particle Swarm Optimization to enhance walking speed to 1.5 m/s in a stable manner.
- Authored seven technique reports documenting experimental design, model development, and iterative findings, maintaining clear records for reproducibility and team collaboration.

**Study on the Secondary Damage and Failure Behavior of the Bonding Interface After Repairing the Debonding of CRTS II Type Slab Track Mortar Layer.**

Sep. 2020 - May. 2021

Funded by Student Innovation Training Program, Tongji University

Advisor: Prof. Xu Yude, College of Transporting Engineering, Tongji University

- Using Abaqus to simulate to model the damage process of track panels (crushing and deformation displacement) over time, temperature, and humidity with the CRTS II China's high-speed rail track model.
- Discovered weakness points of the panels at several locations. Analyzed destruction trends over time, environmental parameters at critical damage points. Suggested parameters for further optimization.

**Optimization Design for Express Bus Stops Stations.**

Mar. 2019 - Sep. 2020

Funded by Student Innovation Training Program, Tongji University

Advisor: Prof. Teng Jing, College of Transporting Engineering, Tongji University

- Proposed and led a project inspired by the discrepancy between scheduled and actual wait times on Dalian City Bus No.505, addressing urban public transportation inefficiencies.
- Designed and distributed questionnaires and conducted field observations, collecting data on stop intervals and passenger boarding and alighting patterns across multiple bus rides.
- Developed and applied a Maximum Likelihood Estimation (MLE) model to analyze collected data and generate optimized bus stop scheduling to reduce passenger wait times.
- The launch of new bus line is expected to save 520 million dollars per year.

## LEADERSHIP

### Teaching Assistance

La Jolla, CA, US

Under Department of Cognitive Science

Apr. 2025 - Jun. 2025

- Course: DSGN 1 (Design of Everyday Things)
- Practiced English communication skills and helped balance the relationship between students and lecturer.

### Team Leader of Chassis Group

Shanghai, China

Tongji University ZEAL Eco-Power

Sep. 2020 - Dec. 2021

- Participated in two chassis projects, designed, machined and assembled the prototype car Z201 and Z202.
- Researched, designed, and manufactured hydrogen fuel system for the first time, completing road tests.

### Self-Driving Project

Under instruction of Professor Huang Yanjun, School of Automotive Studies, Tongji University

- Led a team to build an environment, design tests, and simulate vehicle locomotion
- Independently mastered Carla, an open-source self-driving platform, from the ground up
- Successfully deployed a compact obstacle detection computer vision model on Raspberry Pi 4b

## EXTRACURRICULAR EXPERIENCE

### Volunteer

Shanghai, China

Tongji University Student Union

Oct. 2021 - Nov. 2021

- Core staff in 2021 China International Import Expo, hosting provincial ministerial-level leaders.
- Planning a customized exhibition route for every recipient, accompanying them throughout the process and preparing for emergency response.
- Responsible for coordinating the exhibitor department, the organization department, and logistics department during the tour.

### Advanced Open Water Diver

Jul. 22nd 2023

- Specialty: Elephant Sealing, Underwater Navigating, Nitrox Diving, Deep Diving

### Marathon Runner

- Berkeley Half Marathon, 2:08:38.

Nov. 23rd 2023

## AWARDS

- Won the 3rd Place in the 3rd Shell Eco-marathon China Station (UC Z201) Nov. 2021
- Won the 4th Place in the 3rd Shell Eco-marathon China Station (Prototype Car Z202) Nov. 2021
- Received the Third-Class Scholarship of Outstanding Student at Tongji University 2021
- Received the Third-Class Scholarship of Outstanding Student at Tongji University 2020
- Received the First-Class Scholarship of Outstanding Student at Tongji University 2019

## SKILLS

- **Programming Languages:** Python, MATLAB, SQL, Java, C/C++, Assembly, Golang
- **Libraries & Framework:** PyTorch, Numpy, CVXPY, Scikit-learn, Pandas, Carla, MNE
- **Developer Tools:** Anaconda, Github, Docker, Kubernetes
- **Language Skills:** Mandarin (Native), English (IELTS 7.5), German (CEFR C1)